Lab Worksheet	Identifying Schools Near Rivers Using GeoPandas
Duration	**50 minutes**
Objectives:	**In this hands-on lab, learner will:** • **Load vector datasets using GeoPandas.** • **Explore the datasets.** • **Create a 500 m buffer around rivers.** • **Identify schools located within the river buffer.** • **Visualize the results.**

Instruction

1. Use the dataset provided to complete the lab tasks.
2. Follow the task sequence to complete the tasks.
3. Answer the questions in each task.

Task 1: Load the datasets

Load the two shapefiles and display:

- Number of features
- Coordinate Reference System (CRS)

Code:

```
import geopandas as gpd

schools = gpd.read_file("schools.shp")
rivers = gpd.read_file("rivers.shp")

print("Schools:", len(schools))
print("Rivers:", len(rivers))

print(schools.crs)
print(rivers.crs)
```

Expected Output
Schools: 120
Rivers: 8

CRS: EPSG:4326

Task 2: Reproject the datasets

Generate a buffer and plot the result.

Code:

```
schools = schools.to_crs(epsg=3857)
rivers = rivers.to_crs(epsg=3857)
```

Question: *Why is reprojection necessary before creating a buffer?*

Task 3: Create a 500 m buffer around rivers

Generate a buffer and plot the result.

Code:

```
buffer = rivers.buffer(500)

buffer_gdf = gpd.GeoDataFrame(
    geometry=buffer,
    crs=rivers.crs
)

ax = rivers.plot(figsize=(8,6), color="blue")
buffer_gdf.plot(ax=ax,
                color="lightblue",
                alpha=0.4)
```

Task 4: Identify schools inside the buffer

Use a spatial join to identify schools located within 500 metres of a river.

Display the total number of affected schools.

Code:

```
affected = gpd.sjoin(
    schools,
    buffer_gdf,
    predicate="within"
)

print("Schools within 500 m:",
      len(affected))

affected.head()
```

Task 5: Refelction

Q1. Which GeoPandas function creates a buffer?

Q2. Which function performs a spatial join?

Q3. Why do we use a spatial join?



You have completed the tasks.